



Installation Manual

NordBase Small Foundation





INTRODUCTION

NordBase Foundation System

NordBase is a concrete-free steel foundation system for EV charger and bollard installations. Bolt-together assembly, no cure time, no rebar — ready to load the moment it's backfilled.

NORDINFRA

Rev. 1.1 — 2026-08-24

INTRODUCTION

NordBase Foundation

NordBase is a modular steel foundation for EV charging equipment and bollards — engineered to replace poured concrete. Made from ZAM-coated steel using kerf-fold assembly, it's bolt-together rather than cast: no cure time, no on-site welding. One foundation platform, one shared bolt pattern, for HPC, DC, AC chargers and bollards.

TOOLS AND MATERIALS REQUIRED



Wrench / torque tool



PPE



Leveling tool



Compactor



Shovel

SCOPE OF DELIVERY

- Foundation body — pre-assembled steel shell with factory-installed stud bolt kit on the top plate
- Hardware kit — nuts, washers, and stud connection, factory-installed — no loose hardware bags to lose on site
- Adapter plate or bollard base plate — model-specific, bolts directly to the foundation's top plate pattern





OVERVIEW

NordBase Foundation System

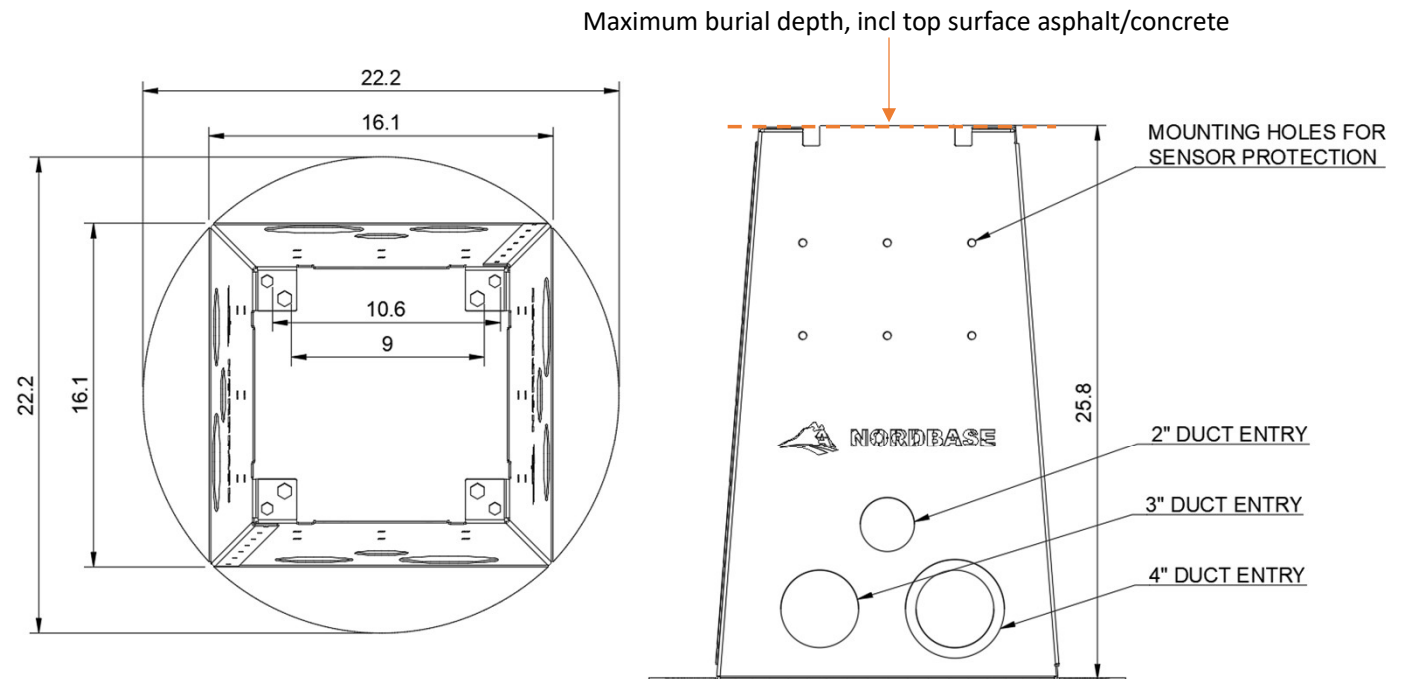
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PRODUCT OVERVIEW

NordBase Small



- MATERIAL: STEEL, SS GRADE 33
- COATING: ZM115 (ZINC-ALUMINUM-MAGNESIUM) — SELF-HEALING CORROSION PROTECTION AT CUT EDGES
- APPLICABLE STANDARDS: ASTM A1046, ASCE 7-22, IBC 2021, NEC, ASTM A500
- STEEL DOMESTICALLY MELTED & MANUFACTURED IN THE USA — BUY AMERICA / BABA COMPLIANT



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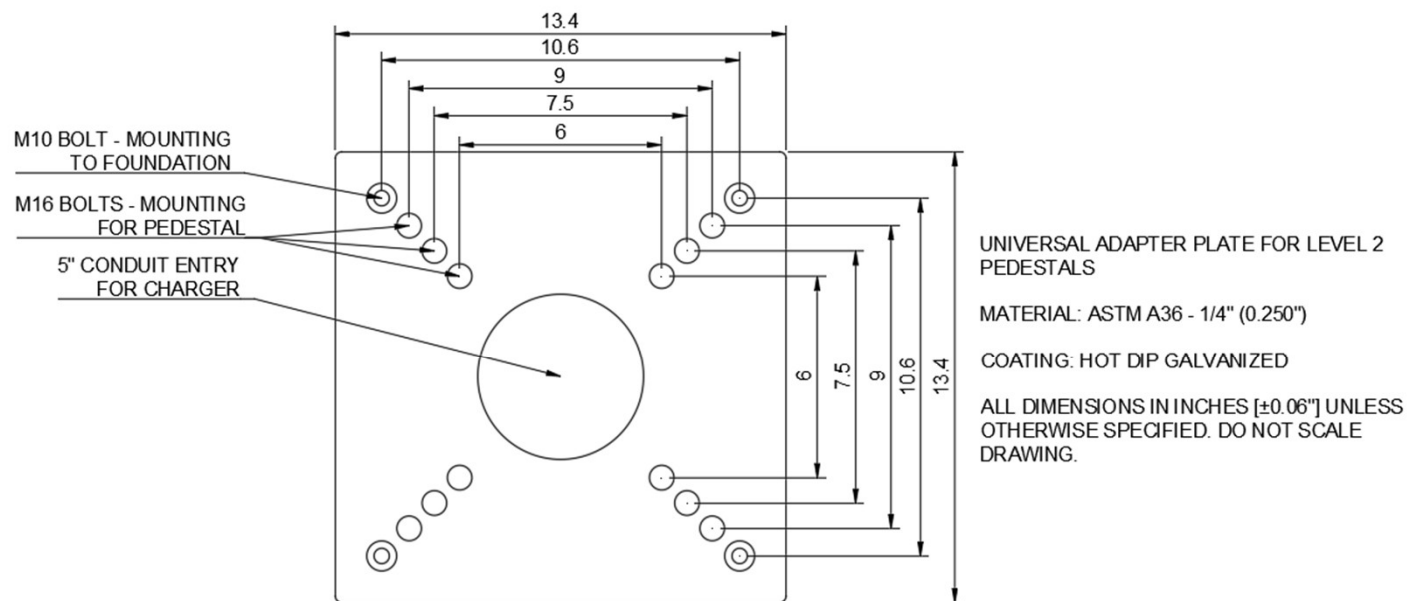
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PRODUCT OVERVIEW

NordBase Small – Adapter Plate



The included universal adapter plate covers 6", 7.5", 9" bolt patterns, matching most Level 2 pedestal chargers on the US market. Model-specific adapter plates are also available for pedestals outside this range — contact Nordinfra to confirm compatibility with your specific charger model



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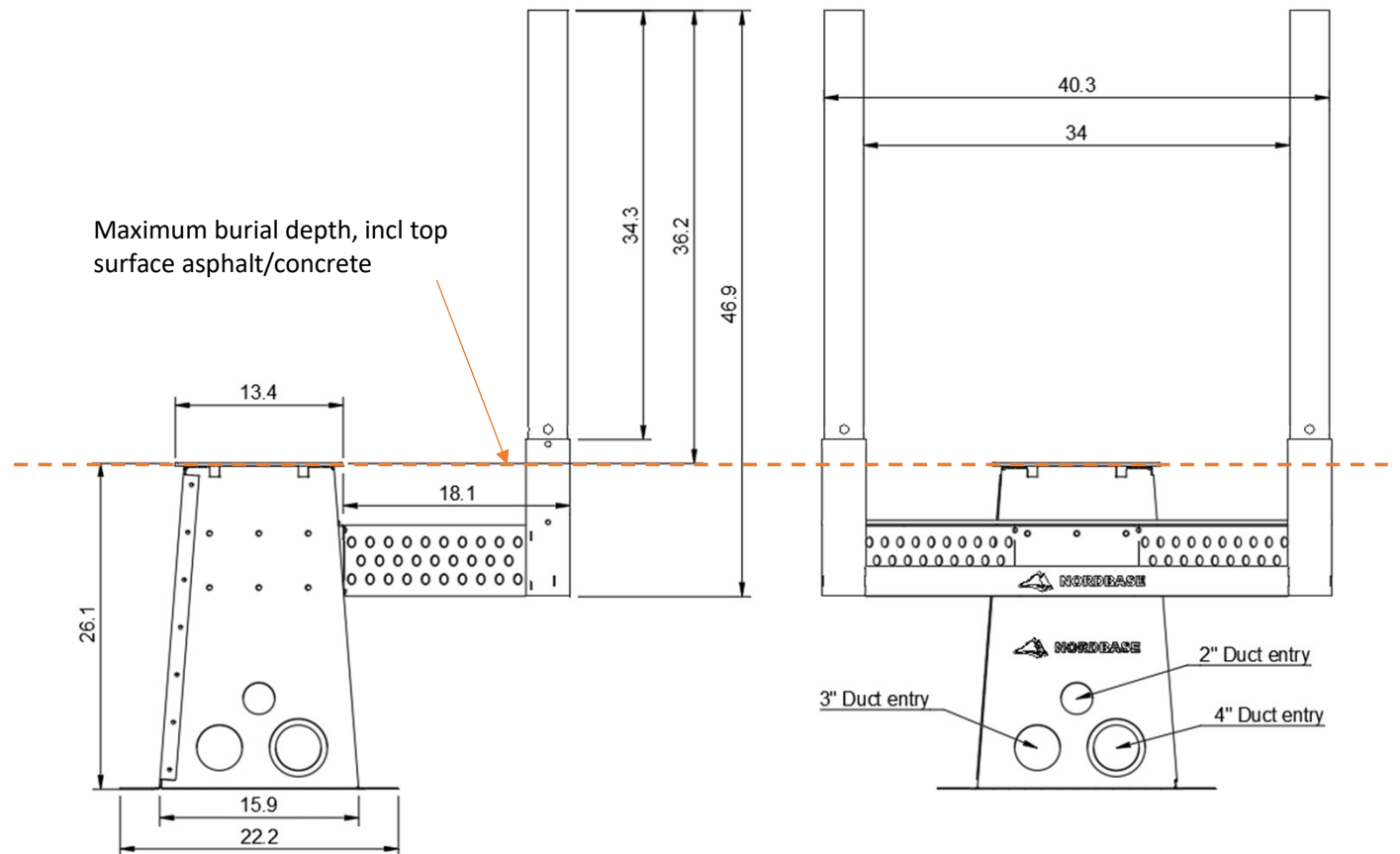
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PRODUCT OVERVIEW

NordBase Small – Sensor Pole





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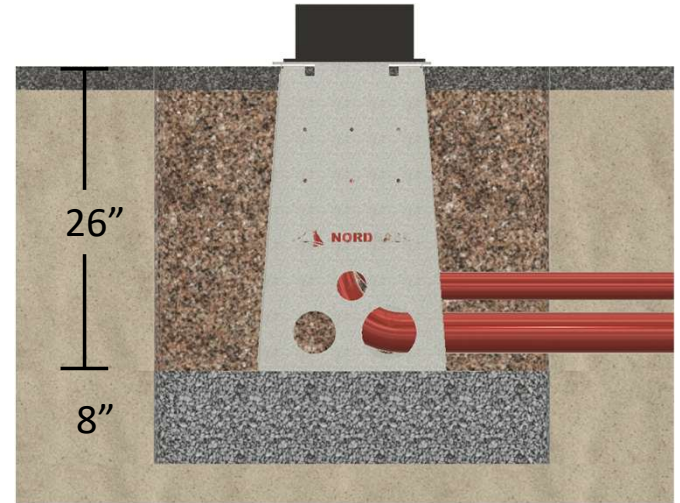
1

Prepare the Trench



Step 1.1: Excavate the Trench

Excavate to the required depth and width for the foundation being installed — width + 10". Lay a 8" layer of compacted base course / crushed stone at the bottom of the trench before placing the foundation.



Step 1.2: Position the Foundation

Place the foundation on the compacted base course. Align conduit openings with the planned cable routing, then route conduit through the entry point before backfilling.



Call 811 (or your local utility notification center) at least 2-3 business days before excavating. Confirm the area is free of underground utilities before digging.

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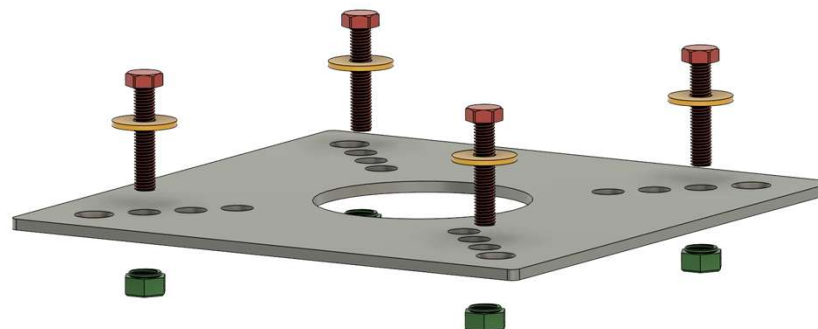
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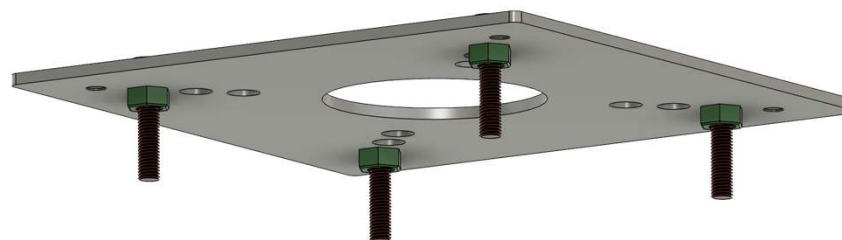
2 Prepare Adapter Plate



Align Bolts with self clinching nuts



Tighten bolts with impact wrench



REFERENCE POINTS

Step 2.1: Self clinching nuts

NOTE: Only for 7,5" and 6" hole pattern. Align self-clinching nuts with required holes on under side of adapter plate.

Step 2.2: Mounting Hardware

Mount M12 bolts on self-clinching nuts. Use impact wrench tighten until the surface of the nut is flush with adapter plate. Approximately 80-100 Nm

NOTE

Mounting self-clinching nuts can be done prior or after installation of foundation.

If you are uncertain of hole pattern — contact your supplier of pedestal for more information.

If self-clinching nuts are mounted in the wrong holes, simply attach bolt and use hammer to remove. Nuts can be used 2 times.

DON'T attempt to insert fastener with a hammer blow — under any circumstances. A hammer blow won't permit the sheet metal to flow and develop an interlock with the fastener's contour.



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3 Backfill and Compact



REFERENCE POINTS

Step 3.1: Adapter Plate

Install adapter plate on foundation prior to back-fill to stabilize the base.

Step 3.2: Bracing

Before backfilling, temporarily brace or support the foundation to prevent movement during compaction. Confirm the foundation is both level and plumb (vertically straight) — check both before proceeding.

Step 3.3: Surround Backfill

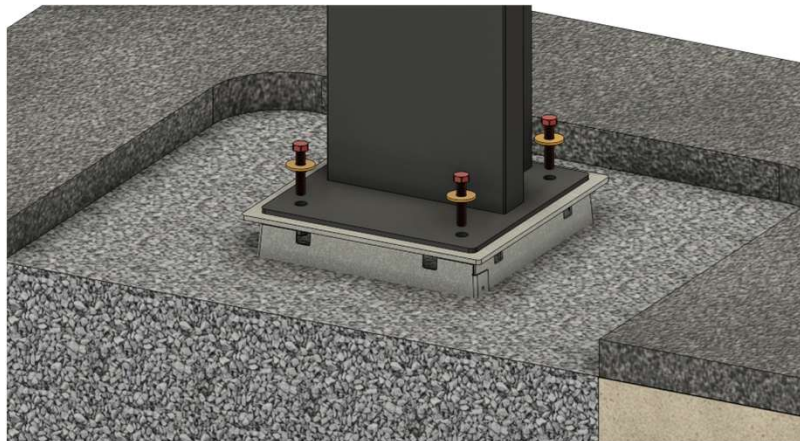
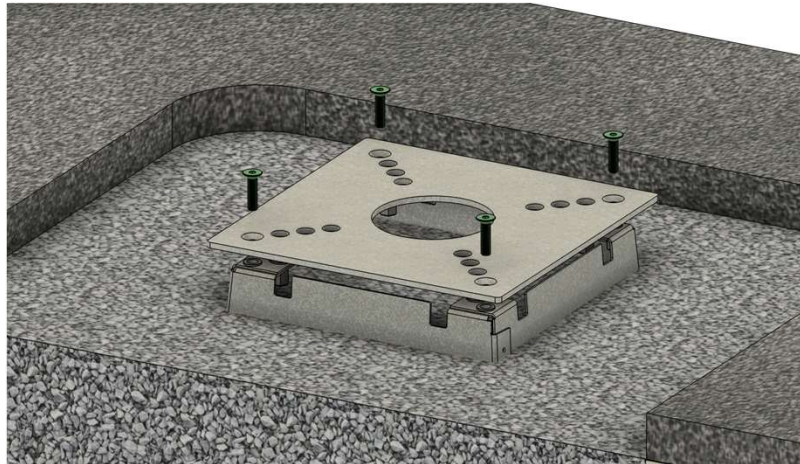
Crushed stone compacted in layers within the 200mm (8") zone around the foundation. Provides free drainage and lateral stability.

Step 3.4: Mounting Hardware

Nuts and washers secure the pedestal or bollard base plate directly to the foundation's bolt pattern.

Instructions

Backfill in 150-200mm (6-8") lifts, compacting thoroughly after each layer. Re-check the foundation with a spirit level after each lift. Confirm the top plate is level and flush with finished grade before final tightening.





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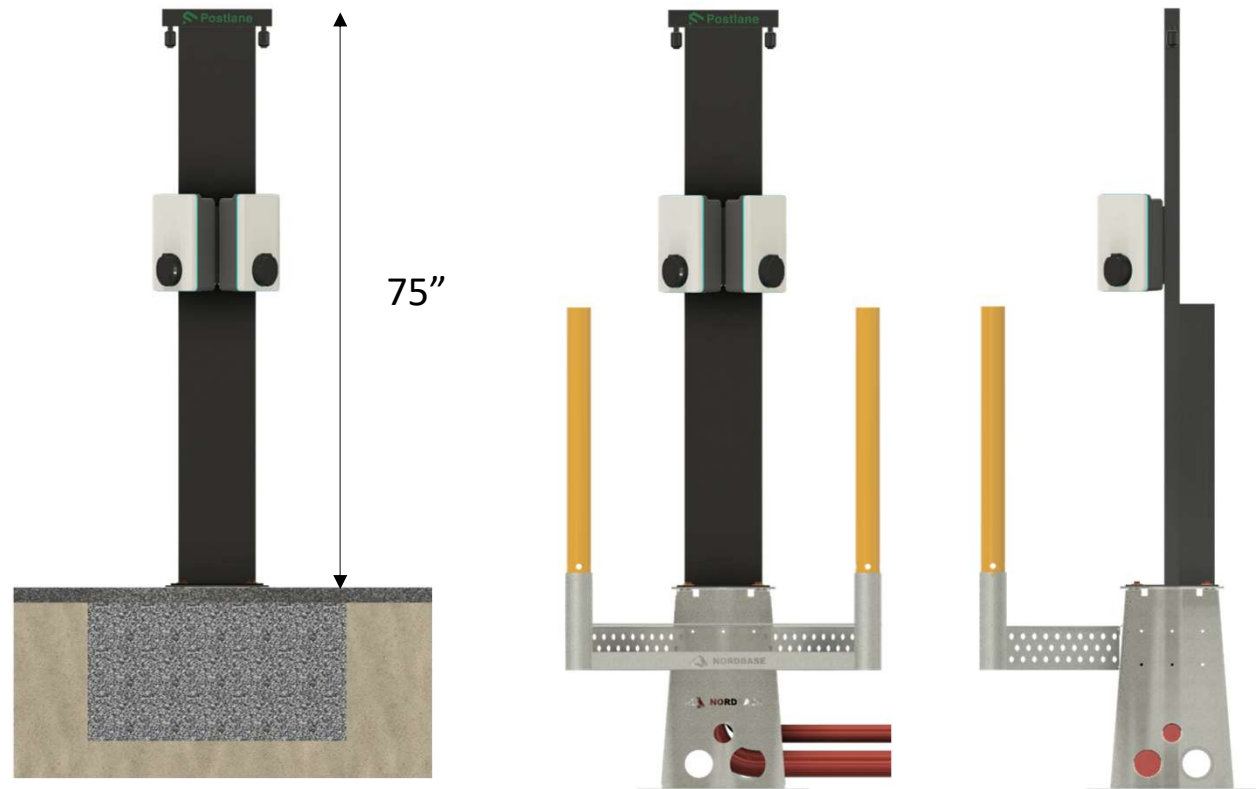
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TECHNICAL REFERENCE

Installed product



The Nordbase Small foundation is structurally verified per ASCE 7-22 / IBC 2021 for a charger up to 10" (W) × 10" (D) × 75" (H), 300lbs, under 130 mph basic wind speed and SDS = 1.5g seismic design category, when installed per the backfill requirements in this manual (≥95% Proctor compacted, free-draining gravel, min. 16 mm). Governing load case: wind overturning, DCR = 76.5% (pass).



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PRODUCT LINEUP

Accessories

Compatible hardware and covers for foundation and bollard installations.



Bollard

Sch. 10 (standard) / Sch. 40 (heavy-duty)

36" x 4.5" OD, ASTM A500 Gr. B
Powder coat / HDG + powder coat



Bollard Base Cover

14 GA carbon steel, spun

Ø11.4", Ø4.6", 2,6"
Powder coat, OSHA Safety Yellow



Bollard Foundation

SS Grade 33 + ZM115-coating

5/8"-11, HDG per ASTM A153
No loose hardware on site

Bolt kit uses the foundation's factory-installed top plate pattern. Base cover slides over the bollard pipe to conceal mounting hardware.



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PRODUCT LINEUP

Accessories

Compatible hardware and covers for foundation and bollard installations.



Sensor Pole

Mounts directly to Nordbase Small, Medium and Large foundation.

Poles: Stainless steel 304, Powder coat, OSHA Safety Yellow

Frame: Hot Dip Galvanized

*Sensor Pole frame is fabricated from formed steel tube, not Schedule 10/40 pipe, and is not rated or tested for vehicle impact. For collision protection applications, use the NordBase Bollard (Sch. 10/40, ASTM A500 Gr. B).

Bolt kit uses the foundation's factory-installed top plate pattern. Base cover slides over the bollard pipe to conceal mounting hardware.



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REFERENCE

General Recommendations

FINAL INSPECTION CHECKLIST

Before proceeding to pole or bollard mounting, confirm:

- ☒ Backfill material is properly compacted in all layers
- ☒ No voids or gaps remain around the foundation
- ☒ Top plate is flush with finished grade and aligned with the project layout
- ☒ Foundation is level and plumb
- ☒ Foundation is ready for hardware installation

ROW ALIGNMENT

For multiple installations in a row (e.g. parking lot bollards or charger banks), align each foundation's bolt pattern parallel to adjacent units and to any marked reference lines before backfilling.

SITE CONDITIONS

If excavation encounters bedrock, a high water table, or other unexpected ground conditions, consult a geotechnical engineer before proceeding. Standard installation depths assume typical soil conditions as outlined in this manual.

STORAGE BEFORE INSTALLATION

Store foundations on a flat, dry surface prior to installation. Avoid prolonged ground contact in standing water. Factory-installed hardware should remain covered or protected from debris until final assembly.

This manual provides general installation guidance based on standard practice and is not a substitute for project-specific engineering. The installing contractor is solely responsible for excavation dimensions, bedding preparation, foundation orientation, backfill material and compaction, conduit installation, and correct equipment mounting.



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TECHNICAL NOTES

Material & Compliance

COATING & CORROSION

NordBase foundations are fabricated from SS Grade 33 steel with a ZM115 (Zinc-Aluminum-Magnesium) coating. The ZM115 coating provides self-healing corrosion protection at cut edges — the zinc-aluminum-magnesium alloy sacrificially protects exposed steel even where the coating has been cut or scratched, unlike standard galvanizing. Minor surface rust may appear over time; this is cosmetic only and will not spread or affect structural performance.

Steel is domestically melted and manufactured in the USA, supporting Buy America, Buy American (BABA) requirements for federally funded infrastructure projects, including NEVI-funded EV charging installations.

APPLICABLE STANDARDS

- ASCE 7-22
- IBC 2021
- ASTM A1046
- NEC — National Electrical Code
- ASTM A500

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NORDBASE

Safety Information



Personal Safety

PPE requirement text — gloves, safety glasses, steel-toe boots.



Proper Handling

Handling guidance text for tools and components.



Public Safety

Guidance on protecting the public — bolt caps, barriers.



Utility Locates

Call 811 (or your local utility notification center) at least 2–3 business days before excavating. Have all underground utilities marked before digging.



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*This manual is a recommendation based on standard practice.
Final geotechnical suitability must be verified by the installer.*

